

◦ 18.06 Linear Algebra - Lecture 7 to 9
Textbook Problem Sets

◦ Problem Set 3.2 pg 47

1 a) $A = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 1 & 2 & 3 & 6 & 9 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ ✓
($m \times n = (3 \times 5)$)

$R(A) = 2$ ✓

Free Variables = $n - r = 5 - 2 = 3$ ✓ $\{x_2; x_4; x_5\}$

Pivot Variable = $r = 2$ $\{x_1; x_3\}$

b) $B = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 4 & 4 \\ 0 & 8 & 8 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 4 & 2 \\ 0 & 4 & 4 \\ 0 & 0 & 0 \end{bmatrix}$ ✓

Pivot Variables = $r = 2$ $\{x_1; x_2\}$ ✓

Free Variables = $n - r = 3 - 2 = 1$ $\{x_3\}$. ✓

3) a) $U = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = R$

b) $U = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 4 & 4 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 4 & -2 \\ 0 & 4 & 4 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = R$

↳ Nullspace True $\rightarrow$ Nullspace remains the same from A to U and from U to R .

2 a) ~~$R = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ set $x_2 = 1$ $Rx = 0$, set $x_4, x_5 = 0$~~

~~$\therefore x_1 + 2(1) + 2x_3 + 4(0) + 6(0) = 0$~~

~~$x_1 + 2x_3 = 0$ ①~~

~~$0 + 0 + x_3 + 2(0) + 3(0) = 0$~~

~~$\therefore x_3 = 0 \therefore x_1 = 0$~~

$$2a) R = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{free: } x_2; x_4; x_5$$

set $x_2 = 1, x_4 = 0, x_5 = 0$ for $Rx = 0$

$$x_1 + 2(1) + 0 + 0 + 0 = 0 \Rightarrow x_1 = -2$$

$$x_3 + 2(0) + 3(0) = 0 \Rightarrow x_3 = 0$$

$$x_2: \text{special solution} = \{-2 \ 1 \ 0 \ 0 \ 0\}$$

set $x_4 = 1, x_2, x_5 = 0$:

$$x_1 + 2(0) + 0 = 0 \Rightarrow x_1 = 0$$

$$x_3 + 2(1) + 3(0) = 0 \Rightarrow x_3 = -2$$

$$x_4: \text{special sol} = \{0 \ 0 \ -2 \ 1 \ 0\}$$

set $x_5 = 1, x_2, x_4 = 0$

$$x_1 + 2(0) + 0 + 0 + 0 = 0 \Rightarrow x_1 = 0$$

$$x_3 + 2(0) + 3(1) = 0 \Rightarrow x_3 = -3$$

$$x_5: \text{special solution} = \{0 \ 0 \ -3 \ 0 \ 1\}$$

$$b) R = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{free} = \{x_3\}$$

set $x_3 = 1$

$$x_1 - (1) = 0 \Rightarrow x_1 = 1$$

$$x_2 + (1) = 0 \Rightarrow x_2 = -1$$

$$x_3 \rightarrow \text{special solution} = \left\{ 1 \ \underset{x}{-2} \ 1 \right\}$$

- 5) a) False $\rightarrow$ any singular square matrix has free variables
 b) True $\rightarrow$ invertible matrix cannot have free variables
 c) True $\rightarrow$ matrix rank r which is # pivot variables n columns to hold pivots cannot exceed n .
 d) False $\rightarrow$ [x] True $\rightarrow$ only m rows to hold pivots
 $\rightarrow$ If can be inverted $\rightarrow$ no free variables.

7) a) $U = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

$U = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

b) $U = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

* $\rightarrow$ Only drop a row on a new pivot. $x_2 \rightarrow$ row 1
 $x_4 \rightarrow$ row 2
 $x_5 \rightarrow$ row 3 } done

9) $3 \times 5 \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$

$\Rightarrow$ The x_5 is a free variable.

set $x_5 = 1$

$x_1 + (0)x_2 + (0)x_3 + (0)x_4 + (1) = 0$

$x_1 = -1$

Special Solution = $\{-1, 0, 0, 0, 1\}$

11) 5 pivots
 5 pivots

$\rightarrow$ Why? ~~Only~~

$\rightarrow$ Only invertible matrices have Nullspace $x=0$
 $\rightarrow$ invertible matrices have 0 free var so 5 pivots

$\rightarrow$ 5 pivots mean that 5 axes are required to span the vector space.

$\rightarrow$ Reasoning here works

$\rightarrow$ bec can solve $Ax=b$ and every b is in the column space.

Idea is that bec 0 free then spanning R^5 so every b in R^5 is solvable

$\Rightarrow$ Sol describe result not cause.

13) $x - 3y - z = 12$ parallel $x - 3y - z = 0$
 $\hookrightarrow 1 \times \text{eqn}$ $\hookrightarrow \text{Nullspace}$

$\hookrightarrow \text{Rank } r = 1$

$\hookrightarrow \text{Free variables} = n - r = 3 - 1 = 2 \rightarrow y, z$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \underbrace{\begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix}}_{x_p} + y \underbrace{\begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}}_{x_n} + z \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Set $y=1, z=0$:

$$x - 3(1) - (0) = 0$$

$$x = 3$$

$\therefore y \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} ; z \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

Set $z=1, y=0$:

$$x - 3(0) - (1) = 0$$

$$x = 1$$

* $\rightarrow$ Think 4 is a typo
 $\hookrightarrow$ Cannot figure out why it would be 4.

15) $N(A) = \text{all combinations } (2, 2, 1, 0) \text{ \& } (3, 1, 0, 1)$

$$A \begin{bmatrix} 2 \\ 2 \\ 1 \\ 0 \end{bmatrix} = 0 \text{ \& } A \begin{bmatrix} 3 \\ 1 \\ 0 \\ 1 \end{bmatrix} = 0 \quad \{Ax=0\} \{Rx=0\}$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$S_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \\ 0 \end{bmatrix} \quad S_2 = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 1 \end{bmatrix} \rightarrow \text{pivots} = \{x_1, x_2\}$$

$\hookrightarrow$ Know how pivots work

$\therefore A =$

$r=2 \quad n-r=2 \quad \therefore n=4 \Rightarrow m \times n = 2 \times 4$

so $A_{m \times n} \Rightarrow A_{2 \times 4}$ as simplest to accommodate
 4 rows with 2 pivots ✓

$\therefore \text{row} = \{a \ b \ c \ d\}$

$$2a + 2b + 1c + 0d = 0$$

$$3a + 1b + 0c + 1d = 0$$

$$A = \begin{bmatrix} 1 & 0 & -2 & -3 \\ 0 & 1 & -2 & -1 \end{bmatrix}$$

$\hat{R} = \begin{bmatrix} 1 & 0 & c_1 & d_1 \\ 0 & 1 & c_2 & d_2 \end{bmatrix}$

$A = \begin{bmatrix} 1 & 0 & -2 & -3 \\ 0 & 1 & -2 & -1 \end{bmatrix}$ ✓

Each row

For S_1 $\{c_1, c_2\}$ remove $\{1, 0\}$ and $\{0, 1\}$

For S_2 $\{d_1, d_2\}$ remove $\{1, 0\}$ and $\{0, 1\}$.

17) $C(A)$ contains $(1, 1, 5)$; $(0, 3, 1)$
 $N(A)$ contains $(1, 1, 2)$
 $S_1 = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \right\}$

$R_x = 0$ $R \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = 0$

$r = \text{rank } 3 \times 2 \quad (3 \times 3)(3 \times 1) = 0 \quad r = 2 \quad 2 \times 3$
 $n - r = 1/0 \quad n - r = 1$

~~$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$~~ $R = \begin{bmatrix} 1 & 0 & -1/2 \\ 0 & 1 & -1/2 \end{bmatrix}$

~~$A = \begin{bmatrix} 1 & -0 \\ 1 & -3 \\ 5 & -1 \end{bmatrix}$~~

~~$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 3 \\ 0 & 5 & 1 \end{bmatrix}$~~

~~$A = \begin{bmatrix} a & 1 & 0 \\ b & 1 & 3 \\ c & 5 & 1 \end{bmatrix}$~~

~~$a + 1 + 0 = 0 \quad a = -1$
 $b + 1 + 6 = 0 \quad b = -7$
 $c + 5 + 2 = 0 \quad c = -7$~~

~~$A = \begin{bmatrix} -1 & 1 & 0 \\ -7 & 1 & 3 \\ -7 & 5 & 1 \end{bmatrix}$~~

Yes
 Ensure
 solution
 are in $\text{Col}(A)$

Let's calculate

$A = \begin{bmatrix} 1 & 0 & a \\ 1 & 3 & b \\ 5 & 1 & c \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & a \\ 1 & 3 & b \\ 5 & 1 & c \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
 $3 \times 3 \quad 3 \times 1$

$(1)(1) + (0)(1) + 2a = 0 \quad a = -1/2$
 $(1)(1) + (3)(1) + 2b = 0 \quad b = -2$
 $(5)(1) + (1)(1) + 2c = 0 \quad c = -3$

$A = \begin{bmatrix} 1 & 0 & -1/2 \\ 1 & 3 & -2 \\ 5 & 1 & -3 \end{bmatrix}$

$\checkmark \rightarrow A = \begin{bmatrix} 1 & 0 & -1/2 \\ 1 & 3 & -2 \\ 5 & 1 & -3 \end{bmatrix}$

20) $N(A) = C(A) \quad A_{m \times n} = A_{2 \times 2}$

$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$2 \times 2 \quad X = 0$

$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$n = 2 \quad r = 1$
 $n - r = 1$

$R_x = 0 \quad \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$

No because $x = c \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Instead $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

$C(A) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$C(A) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$N(A) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

25) $N(A) = C \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}$ rank = 2

~~R/R~~ $S_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}$ $S_2 = \begin{bmatrix} 6 \\ 3 \\ 0 \\ 3 \end{bmatrix}$
 $n \times m = 4 \times 1$

$R = \begin{bmatrix} 1 & 0 & a_1 & b_1 \\ 0 & 1 & a_2 & b_2 \end{bmatrix}$

$m \times n$
 $(4 \times 4)(4 \times 1)$

$(1)(2) + (0)(1) + a_1(0) + b_2(1) = 0$

$b_2 = -2$

$(0)(2) + (1)(1) + a_2(0) + b_2 = 0$

$b_2 = -1$

$\therefore R = \begin{bmatrix} 1 & 0 & a_1 & -2 \\ 0 & 1 & a_2 & -1 \end{bmatrix}$

2×4

where $\begin{cases} a_1 = \text{any } \mathbb{R} \\ a_2 = \frac{1}{2}a_1 \end{cases}$ X

1 First we know $n=4$

2 ~~$S_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}$~~ so x_2, x_3, x_4 could be free variables

3 We know ~~$r \leq 2$~~ $r \leq 3$ as both x_2 and x_4 cannot be free

4 ~~$m \times n$~~ We know $m=4$ ~~$(m \times 4)(4 \times 1) = (4 \times 1)$~~ $\therefore m=4$

5 $\therefore r \leq n$ & $r \leq m \Rightarrow$ Either 0 or ∞ solutions
 $\hookrightarrow$ requires $n-r > 1$

6 ~~$m \times n$~~

7 $\therefore r=2$ $\therefore n-r=2 = 4-2$

$R = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Re-attempt:

$\hookrightarrow$ ~~Originally correct~~ $S_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}$ so either 2 or 3 free variable

$\hookrightarrow$ know $n=4$ (also $(m \times 4)(4 \times 1) = (4 \times 1)$ $\therefore m=4$)

$\hookrightarrow$ Nullspace is multiple of 1 vector $\Rightarrow \dim \text{Nullspace} = 1$

$\hookrightarrow$ Free Variables = $\dim \text{Nullspace} = 1$

$\therefore m-r=1$ $n=4$ $\therefore r=3$

$\therefore A = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$



28) Why A and $-A$ has same rref R ?

↳ $A = -1 \cdot A$

↳ Negative A is A scaled by a -1 constant

↳ While reducing the rows of $-A$, the -1 scalar can be multiplied out of every row, resulting in A which reduces to the same R .

↳ Believe this is a reasonable answer ✓

↳ A and $-A$ have same

linear relationships. Scaling A by (-1) to get $-A$ preserves the relationship $\Rightarrow$ Nullspace remains same for both.

34) $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 2 & 4 & 6 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 4 & 6 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow R = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$B = [A \ A] = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 3 \\ 2 & 4 & 6 & 2 & 4 & 6 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
 $\Rightarrow R = \begin{bmatrix} 1 & 2 & 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} = [R_A \ R_A]$

$C = \begin{bmatrix} A & A \\ A & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 3 \\ 2 & 4 & 6 & 2 & 4 & 6 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 4 & 6 & 2 & 4 & 6 \\ 0 & 0 & 3 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
 $\Rightarrow \begin{bmatrix} 1 & 2 & 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow R = \begin{bmatrix} R_A & 0 \\ 0 & R_A \end{bmatrix}$

38) $R = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Set $x_3 = 1, x_4 = 0$:

$x_1 + 2(1) + 3(0) = 0$

$x_1 = -2$

$x_2 + 4(1) + 5(0) = 0 \Rightarrow x_2 = -4$

$S_1 = \begin{bmatrix} -2 \\ -4 \\ 1 \\ 0 \end{bmatrix}$

$S_2 = \begin{bmatrix} -5 \\ 0 \\ 0 \\ 1 \end{bmatrix}$ ✓

Set $x_3 = 0, x_4 = 1$:

~~$x_1 + 0 + 2(0) + 3(1) = 0$~~

~~$0 + x_2 + 4(0) + 5(1) = 0 \Rightarrow x_2 = -5$~~

Set $x_3 = 0, x_4 = 1$:

$0 + x_2 + 4(0) + 5(1) = 0$

$x_2 = -5$

$x_1 + 0 + 2(0) + 3(1) = 0$

$x_1 = -3$

38 con.) $R = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Set $x_1 = 0, x_3 = 1$

$0 + 1x_2 + 2(1) = 0$
 $x_2 = -2$

Set $x_1 = 1, x_3 = 0$

$0 + 1x_2 + 2(0) = 0$
 $x_2 = 0$

$S_1 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} \quad S_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ ✓

41) $A = \begin{bmatrix} 3 & 6 & 6 \\ 1 & 2 & 2 \\ 4 & 8 & 8 \end{bmatrix}; A = uv^T$

$u = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad v = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

$(3 \times 1)(1 \times 3) = (3 \times 3)$

~~$A = \begin{bmatrix} 3 & 6 & 6 \\ 1 & 2 & 2 \\ 4 & 8 & 8 \end{bmatrix}$~~
 $\Rightarrow A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix}$
 $\Rightarrow \begin{bmatrix} 1 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

LD X $\rightarrow$ Use A as is

$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 6 & 6 \\ 1 & 2 & 2 \\ 4 & 8 & 8 \end{bmatrix}$

$u = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad v = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad X \quad u = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix} \quad v = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

43) $A \rightarrow \text{rank } r \quad S \rightarrow (r \times r) \rightarrow \text{invertible}$

$\hookrightarrow$ remove $m-r$ (rows) & $n-r$ (cols) to find S.

$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \end{bmatrix} (2 \times 3) \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 2 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$m-r = 2-2=0$ rows
 $n-r = 3-2=1$ cols.

$S_A = \begin{bmatrix} 1 & 3 \\ 1 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \text{invertible} \quad \checkmark$

$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix} \quad r=3 \quad m=2 \quad n=3 \quad m-r=1 \quad n-r=0$

~~$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$~~ $\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix} \quad r=1 \quad n-r=2 \quad m-r=1$
 $S_{B_{inv}} = \begin{bmatrix} 1 \end{bmatrix}$

43 con.) $C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow r=2 \quad n-r=3-2=1$
 $m-r=3-2=1$

$$S_C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \checkmark$$

45) $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 4 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$P_A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \\ 2 & 7 \end{bmatrix} \quad \& \quad P_A^T = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 6 & 7 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$S_A^T = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \quad \times \quad \text{Fix} \quad S_A^T = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$$

$$S_A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \quad \checkmark$$

$\hookrightarrow S_A$ from original
 P_A^T and not
 $U/R P_A^T$.

46) $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 1.5 & 6 \end{bmatrix} \quad C = \begin{bmatrix} 1 & b \\ c & bc \end{bmatrix}$

$$AB = \begin{bmatrix} 8 & 4 & 16 \\ 16 & 8 & 32 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 8 & 4 & 16 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 1 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow$ Rank 1. $\checkmark$

$$AC = \begin{bmatrix} 1+2c & b+2bc \\ 2+4c & 2b+4bc \end{bmatrix} \Rightarrow \begin{bmatrix} 1+2c & b+2bc \\ 0 & 0 \end{bmatrix} \Rightarrow \text{Rank 1.}$$

$\hookrightarrow$ or Rank 0,
 if $(c=0) \uparrow$

~~305~~

50) $A_{(m \times n)}$ $B_{(n \times n)}$ $AB = I$

Prove $\text{rank}(A) = n$: use $\text{rank}(AB) \leq \text{rank}(A)$.

$$AB = A \cdot B = A_{(n \times n)} \times B_{(n \times n)}$$

$$\therefore \text{AB}_{(m \times n)} = AB_{(n \times n)}$$

$$\text{rank}(AB_{(n \times n)}) \leq n$$

$$\text{as } \text{rank}(AB) \leq \text{rank}(A)$$

$$\therefore \text{rank}(A) = n$$

$\therefore A$ is invertible. ✓

additionally $\left\{ \begin{array}{l} A^{-1}A = I \quad \text{and} \quad BA = I \\ \therefore A^{-1}A = BA \\ \text{B} = A^{-1} \\ \therefore B \text{ is } A\text{'s two sided inverse.} \end{array} \right.$

53) $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 4 \\ 1 & 1 & 8 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 8 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 8 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 2 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ 0 & 3 \end{bmatrix} \quad \checkmark$$

• Problem Set 3.3

1) $A = \begin{bmatrix} 2 & 4 & 6 & 4 \\ 2 & 5 & 7 & 6 \\ 2 & 3 & 5 & 2 \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$

$$\left[\begin{array}{cccc|c} 2 & 4 & 6 & 4 & b_1 \\ 2 & 5 & 7 & 6 & b_2 \\ 2 & 3 & 5 & 2 & b_3 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 2 & 4 & 6 & 4 & b_1 \\ 0 & 1 & 1 & 2 & b_2 - b_1 \\ 0 & -1 & -1 & -2 & b_3 - b_1 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 0 & 1 & 1 & 2 & b_1 - b_3 \\ 0 & 1 & 1 & 2 & b_2 - b_1 \\ 2 & 3 & 5 & 2 & b_3 \end{array} \right]$$

$$\Rightarrow \left[\begin{array}{cccc|c} 2 & 3 & 5 & 2 & b_3 \\ 0 & 1 & 1 & 2 & b_2 - b_1 \\ 0 & 0 & 0 & 0 & b_1 - b_3 - b_2 + b_1 \end{array} \right] \Rightarrow \text{~~matrix~~ \quad \times \quad \rightarrow$$

$$= \left[\begin{array}{cccc|c} 2 & 1 & 3 & 0 & b_3 - 3b_2 + 3b_1 \\ 0 & 1 & 1 & 2 & b_2 - b_1 \\ 0 & 0 & 0 & 0 & -(b_2 + b_3) \end{array} \right]$$

① $\Rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 1 & -2 & (b_3 - 3b_2 + 3b_1)/2 \\ 0 & 1 & 1 & 2 & b_2 - b_1 \\ 0 & 0 & 0 & 0 & -(b_2 + b_3) \end{array} \right]$

② Last equation shows $\therefore -(b_2 + b_3) = 0 \Rightarrow -b_2 - b_3 = 0$
 $\Rightarrow b_2 = -b_3$

$Ax = b$ has a solution when

$$\rightarrow \left[\begin{array}{cccc|c} 2 & 4 & 6 & 4 & b_1 \\ 0 & 1 & 1 & 2 & b_2 - b_1 \\ 2 & 3 & 5 & 2 & b_3 \end{array} \right] \Rightarrow \left[\begin{array}{cccc|c} 2 & 4 & 6 & 4 & b_1 \\ 0 & 1 & 1 & 2 & b_2 - b_1 \\ 0 & -1 & -1 & -2 & b_3 - b_1 \end{array} \right] \Rightarrow \left[\begin{array}{cccc|c} 2 & 4 & 6 & 4 & b_1 \\ 0 & 1 & 1 & 2 & b_2 - b_1 \\ 0 & 0 & 0 & 0 & b_3 + b_2 - 2b_1 \end{array} \right]$$

② Solution to $Ax = b$ where $\therefore$
 $b_3 + b_2 - 2b_1 = 0$

③ $\text{Col}(A)$ is all combinations of $\therefore \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}; \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix}$

④ ~~Special Solutions~~ Special Solutions $\therefore$

Set $x_3 = 0, x_4 = 1 \therefore$ $x_2 + (0) + 2(1) = 0 \quad x_2 = -2$
 $2x_1 + 4(-2) + 6(0) + 4(1) = 0 \quad x_1 = 2$

Set $x_3 = 1, x_4 = 0 \therefore$ $s_1 = \begin{bmatrix} 2 \\ -2 \\ 0 \\ 1 \end{bmatrix}$

$x_2 + (1) + 2(0) = 0 \quad x_2 = -1$
 $2x_1 + 4(-1) + 6(1) + 4(0) = 0 \quad x_1 = -1$

~~Set~~ $s_2 = \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}$

$N(A) \therefore x_{\text{complete}} = c_1 \begin{bmatrix} 2 \\ -2 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}$

$$⑤ [U \ c] \rightarrow [R \ d]$$

$$[U \ c] = \left[\begin{array}{cccc|c} 2 & 4 & 6 & 4 & b_1 \\ 0 & 1 & 1 & 2 & b_2 - b_1 \\ 0 & 0 & 0 & 0 & b_3 + b_2 - 2b_1 \end{array} \right]$$

$$\Rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 2 & b_1/2 \\ 0 & 1 & 1 & 2 & b_2 - b_1 \\ 0 & 0 & 0 & 0 & b_3 + b_2 - 2b_1 \end{array} \right] \Rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 1 & -4 & \frac{1}{2}b_1 - 2b_2 + 3b_3 \\ 0 & 1 & 1 & 2 & b_2 - b_1 \\ 0 & 0 & 0 & 0 & b_3 + b_2 - 2b_1 \end{array} \right]$$

$$\Rightarrow \cancel{[U \ c]} \rightarrow [R \ d]$$

Require a specific solution for c .

$$b = \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix}$$

$$\begin{array}{l} b_1 \\ b_2 - b_1 \\ b_3 + b_2 - 2b_1 \end{array} = 0$$

$$[U \ c] = \left[\begin{array}{cccc|c} 2 & 4 & 6 & 4 & 4 \\ 0 & 1 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$\Rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 2 & 2 \\ 0 & 1 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 1 & -4 & 2 \\ 0 & 1 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$\Rightarrow \left[\begin{array}{cccc|c} 2 & 0 & 2 & -4 & 4 \\ 0 & 1 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$3) A = \begin{bmatrix} 1 & 3 & 3 \\ 2 & 6 & 9 \\ -1 & -3 & 3 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 5 \\ 5 \end{bmatrix}$$

$$\begin{aligned} [U \ c] &= \text{elim}([A \ b]) = \left[\begin{array}{ccc|c} 1 & 3 & 3 & b_1 \\ 2 & 6 & 9 & b_2 \\ -1 & -3 & 3 & b_3 \end{array} \right] \\ &= \left[\begin{array}{ccc|c} 1 & 3 & 3 & b_1 \\ 0 & 0 & 3 & b_2 - 2b_1 \\ 0 & 0 & 6 & b_3 + b_1 \end{array} \right] = \\ &= \left[\begin{array}{ccc|c} 1 & 3 & 3 & b_1 \\ 0 & 0 & 3 & b_2 - b_1 \\ 0 & 0 & 0 & 5b_1 - 2b_2 + b_3 \end{array} \right] \end{aligned}$$

$$\therefore \text{Col}(A) / \text{Solvability condition} : 5b_1 - 2b_2 + b_3 = 0$$

$$5(1) - 2(5) + (5) = 0 \quad \oplus$$

$$\text{Col}(A) = \left\{ \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 \\ 9 \\ 3 \end{pmatrix} \right\}$$

Special Solutions $N(A)$

$$\textcircled{1} \text{ Set } x_2 = 1 \quad 3x_3 = 0 \quad \therefore x_3 = 0$$

$$x_1 + 3(1) + 3(0) = 0 \quad \therefore x_1 = -3$$

$$S_1 = \begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix} \quad N(A) = C_1 S_1 \quad x_n = C_1 \begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix}$$

Particular Solution x_p

$$[U \ c] \Rightarrow C = \begin{bmatrix} b_1 \\ b_2 - 2b_1 \\ 5b_1 - 2b_2 + b_3 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 5 \\ 5 \end{bmatrix}$$

$$c = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$$

$$\text{Set } x_2 = 0 \quad (\text{free var} \rightarrow \text{can be anything} \rightarrow \text{choose simplest}).$$

$$x_1 + 3(0) + 3x_3 = 3 \quad \therefore x_3 = 1$$

$$x_1 + 3(0) + 3(1) = 1$$

$$x_1 = -2$$

$$x_p = \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$$

Additional Info $\rightarrow$ Complete Solution : $x_c = \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} + C_1 \begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix}$ ✓
 $\rightarrow$ Matrix is singular ($r < n$) ($2 < 3$) solvable ($0 = 0$ row in $[U \ c]$)
 b is in the column space ($b = \begin{pmatrix} 1 \\ 5 \\ 5 \end{pmatrix}$ is in $5b_1 - 2b_2 + b_3 = 0$) so in $C(A)$.
 Used free variable $y = 0$.

$$5) \quad A = \begin{bmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 4 & 9 & -8 \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\begin{aligned} [U \mid c] &= \left[\begin{array}{ccc|c} 1 & 2 & -2 & b_1 \\ 2 & 5 & -4 & b_2 \\ 4 & 9 & -8 & b_3 \end{array} \right] \\ &= \left[\begin{array}{ccc|c} 1 & 2 & -2 & b_1 \\ 0 & 1 & 0 & b_2 - 2b_1 \\ 0 & 1 & 0 & b_3 - 4b_1 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & 2 & -2 & b_1 \\ 0 & 1 & 0 & b_2 - 2b_1 \\ 0 & 0 & 0 & b_3 - 4b_1 \end{array} \right] \\ &= \left[\begin{array}{ccc|c} 1 & 2 & -2 & b_1 \\ 0 & 1 & 0 & b_2 - 2b_1 \\ 0 & 0 & 0 & -2b_1 - b_2 + b_3 \end{array} \right] \end{aligned}$$

Solvable Condition : $b_3 - 2b_1 - b_2 = 0$

x_n : special solution (set $x_3 = 1$)

$$x_2 = 0$$

$$x_1 + 2(0) - 2(1) = 0 \quad \therefore x_1 = 2$$

$$x_n = C_1 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

x_p : $C = \begin{bmatrix} b_1 \\ b_2 - 2b_1 \\ b_3 - 2b_1 - b_2 \end{bmatrix}$

Set $x_3 = 0$

$$x_2 = b_2 - 2b_1$$

$$x_1 + 2(b_2 - 2b_1) - 2(0) = b_1$$

$$x_1 = 5b_1 - 2b_2$$

$$x_p = \begin{bmatrix} 5b_1 - 2b_2 \\ b_2 - 2b_1 \\ 0 \end{bmatrix}$$

x_{complete} :

$$\begin{aligned} x_c &= x_p + x_n \\ &= \begin{bmatrix} 5b_1 - 2b_2 \\ b_2 - 2b_1 \\ 0 \end{bmatrix} + C_1 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \end{aligned}$$

10) $A_{2 \times 3}$ with: $x = \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} + c \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$\begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} : 0 \rightarrow 1 \text{ free var}$$

$$c \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} c \\ c \end{pmatrix}$$

13) a) Particular solution x_p is always multiplied by 1.

b) Many particular solutions exist. ~~1~~ ✓
Free variables may be arbitrarily chosen.

c) $\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \end{bmatrix}$; $\begin{bmatrix} 1 \end{bmatrix}$ shorter ($l=\sqrt{2}$) than $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$

d) $\rightarrow$ Homogeneous solution $x_h = 0$ exists.